

DishAgent: Enhancing Dining Experiences through LLM-Based Smart Dishes

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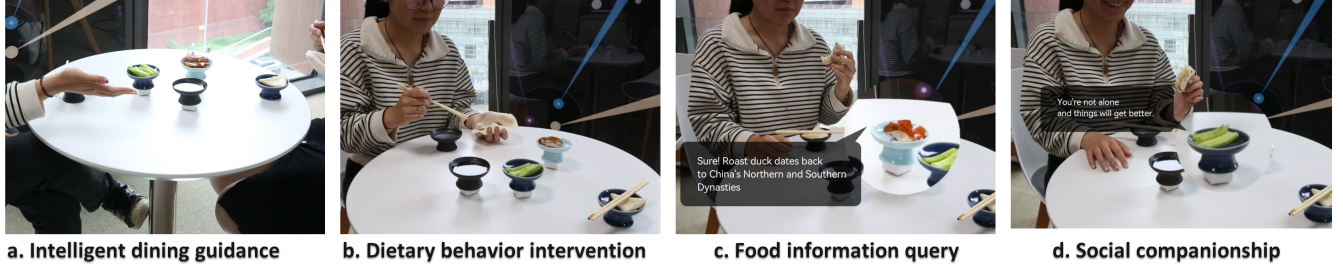


Figure 1: Dining Scenarios.

ABSTRACT

With the rapid advancement of smart technologies, there is an increasing demand to enhance everyday experiences, including dining. Recent Human-Computer Interaction (HCI) research has begun to emphasize the aesthetic, affective, sensual, and sociocultural qualities of directly interacting with food. However, these technologies are often constrained by the material properties of food, limiting their everyday applicability. This research introduces DishAgent, an innovative device equipped with a Large Language Model (LLM)-based smart dish and a swarm robotics system. DishAgent adapts to various dining scenarios by generating appropriate conversational contexts and coordinating the action commands of swarm robots, thereby enhancing the dining experience through real-time interaction. This paper explores the applications of DishAgent in intelligent dining guidance, dietary behavior intervention,

food information query and social companionship, aiming to fill the critical gap in current technologies for simply and intuitively enhancing dining experiences.

CCS CONCEPTS

• Human-centered computing → Ubiquitous and mobile devices.

KEYWORDS

Human Food Interaction; Large Language Model; Smart Dish

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1 INTRODUCTION

With the rapid advancement of smart technologies, there is an increasing demand to enhance everyday experiences, including dining. Previous research has shown that Human-Computer Interaction (HCI) technology in dining can make eating more enjoyable by facilitating interactions between diners and their food, such as letting solo diners connect with other people[14], creating some kind

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of artificial dining companions[1], or contributing to an enjoyment and “reverie” in the eating experiences [13]. However, traditional interactive dining technologies have predominantly focused on visual and auditory enhancements through displays, projections, and sound systems. These approaches often involve high costs and complex setups[7], making them less accessible and detached from the tangible, pleasurable, intuitive aspects of food interaction.

Recent HCI studies have started to emphasize the aesthetic, affective, sensual, and sociocultural qualities of interacting directly with food[6], and focus on the material properties of food and intuitive eating experiences[2, 18]. For example, Deng et.al developed a system for computational food designs that facilitates dynamic dining trajectories via “programming” and “reconfiguring” food items [6]. Nakata et.al proposed a movable, edible robot[15]. Nevertheless, these technologies are often constrained by the material properties of food, limiting their everyday applicability.

Hence, there is a critical gap in creating devices that can implicitly represent food and deliver pleasurable dining experiences simply and intuitively. In this context, dishes—integral to dining and typically blending seamlessly with the food they carry—offer a unique opportunity. Previous studies have demonstrated the potential for dishes to engage in proto-communication with diners using minimal social cues[11].

Building on this potential, this study introduces DishAgent, a smart dish equipped with an LLM-based agent and a swarm robotics system. This system adapts to various dining scenarios by generating appropriate conversational contexts and coordinating the action command of swarm robots. By doing so, we expect that DishAgent can enhance the dining experience through real-time interaction. At the end of the paper, we explored applications in intelligent dining guidance, dietary behavior intervention, food information query and social companionship.

2 RELATED WORKS

2.1 Interactive Dining Tables

Interactive dining table systems show significant potential in enhancing dining experiences and promoting healthy eating. Multimodal interaction interfaces play a crucial role in human-food interaction. For example, Bruijnes et al. demonstrated how interactive mixed reality systems can enhance dining experiences through visual, auditory, and tactile stimuli[3]. Such multimodal stimulation applications help improve users’ perception of food, thereby enhancing the overall dining experience. Additionally, smart dining table systems integrate sensors and control systems to monitor food intake and provide personalized dietary recommendations, as explored by Harper et al.[10]. Cheong et al. designed a system where robots carry trays and collect cutlery, allowing diners to order food via tablets and receive notifications through a coaster pager when their order is ready[5]. The E-Restaurant concept, as described by Kaushal et al, incorporates robotic waiters and an E-Trolley system to serve customers. Cheng et al. designed a smart dining table capable of hosting interactive games and activities in open-plan kitchen/living room settings, further enhancing the dining experience[4]. However, these studies are mainly limited to using devices to enhance the dining experience, without effectively integrating with the food itself.[12]

2.2 Large Language Models (LLMs) for Human Food Interaction

Large language models (LLMs) can understand and generate natural language, enabling the dish to represent food to engage in dialogue with users, providing information about nutrition, cooking methods, and more. Yang et al. proposed ChatDiet, a nutrition-oriented food recommendation framework that combines personal and population models with dynamics to offer personalized and interpretable food recommendations[19]. Additionally, Shi et al. proposed a three-round iterative prompting method to utilize GPT-4 for food effect summarization, showing that 85% of the summaries generated by GPT-4 are factually consistent with the golden reference summary, indicating GPT-4’s great potential in drafting food effect summaries for FDA professionals to review[17]. Rodríguez-de-Vera et al. proposed Dining on Details (DoD), an innovative expert learning framework for food classification. This method cleverly leverages the power of large language models to construct category subsets within the dataset, enabling the identification of meaningful similarities between different food categories[16]. The application of large language models in the food domain primarily focuses on food recommendation and recognition. Research on enhancing the dining experience, particularly on enabling dialogue between humans and food, remains limited.

This paper proposes the DishAgent system, which endows food with mobility through the use of a dining table robot and equips it with the ability to understand and engage in conversation by integrating large language models.

3 DESIGN EXPLORATION

To more effectively explore the innovative dining experiences that smart dinner plates can offer in personalized settings, and to aid the development of future interactive dining table designs, we created DishAgent. This section details the design exploration of the DishAgent, from initial inspiration and hardware system development to interaction design. The overall framework as shown in Fig.2.

3.1 Design Inspiration

Previous studies have suggested that dish robots could engage in primitive communication with diners using minimal social cues[11]. We aim to leverage the perceptual, understanding, and generative capabilities of Large Language Models (LLMs) regarding physical interactions to enhance our DishAgent’s ability to create personalized interactions between humans and food. Specifically, our design goal is for the DishAgent to autonomously perceive social cues in a dining context, providing tangible, pleasurable, and intuitive experiences through interaction.

3.1.1 Perception of Interaction Cues: We envision that DishAgent will be capable of perceiving a variety of interaction cues in everyday dining scenarios, including individual information such as user behavior habits, expressions, and posture; food information involving the type, quantity, and arrangement of food; and interaction information that analyzes how users interact with food, such as eating actions and dining speed.

3.1.2 Generation and Decision-Making for Personalized Interaction: We aim for DishAgent to use Large Language Models (LLMs) to

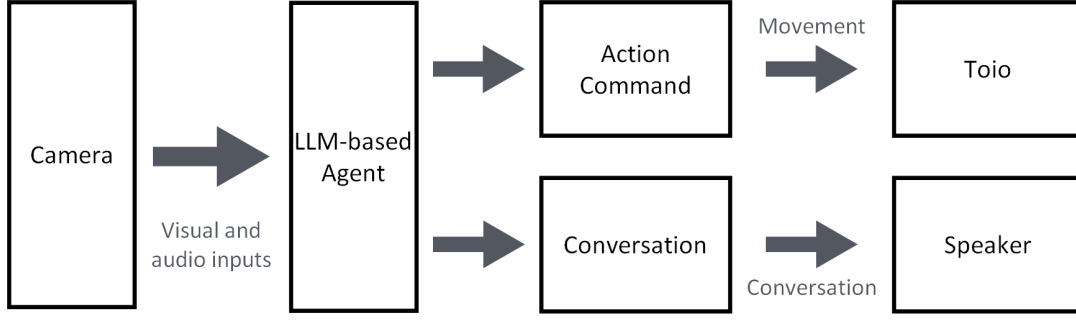


Figure 2: Overall Framework

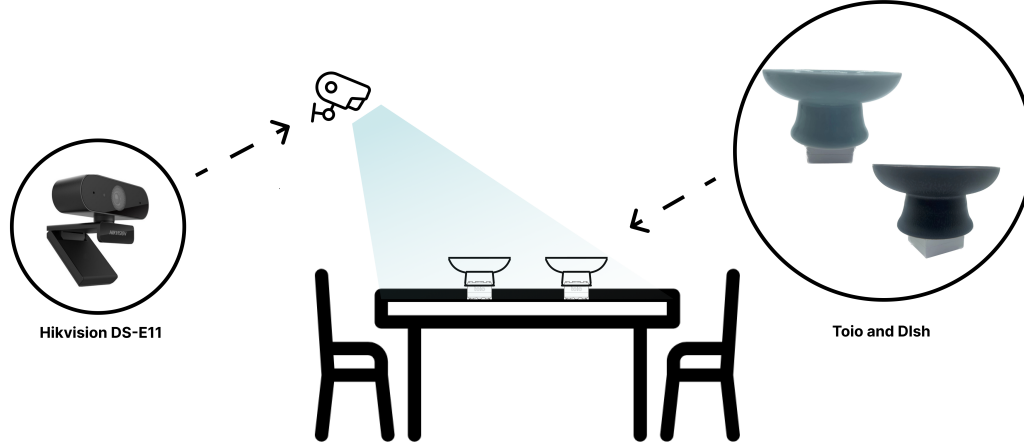


Figure 3: Hardware Design

analyze the collected data and formulate interaction behavioral decisions such as personalized dietary suggestions, emotional conversation or non-verbal expression from food.

3.1.3 Multimodal Interaction Outputs: We intend to equip DishAgent with the capability to dynamically move on the tabletop and interact vocally. Previous research has shown that swarm robots can provide such interactions through an LLM-based agent and can integrate with everyday objects due to their modular nature[8, 9]. Therefore, we combine smart dinner plates with swarm robots (Toio), allowing them to adjust the position of food based on system decisions and provide interactive voice feedback through speakers, offering dietary suggestions and engaging conversations.

3.2 DishAgent System Development

The DishAgent system was developed to include three core modules: input, decision-making, and output (as shown in Fig 3.)

The input module uses a Hikvision desktop camera (model DS-E11, resolution 1920x1080) with an integrated microphone to capture detailed visual and auditory information from the dining environment and user behavior in real time, providing visual and audio inputs to the system.

These data including visual and auditory inputs, are processed in the decision-making module. An LLM-based agent will analyze the information and provide interaction behavior commands through a structured prompt. These commands include a context for conversation and the action command of swarm robots[9].

The output module of the DishAgent system features swarm robots Toio (model TPH-1009C), embedded beneath traditional dishes. These robots precisely control the positioning and movement of the food, tailored to user preferences to enhance the interactive dining experience. Additionally, the DishAgent incorporates a speaker module for auditory feedback, which is connected to an ESP32 module via Bluetooth to facilitate audio communication.

4 APPLICATION

DishAgent enhances the user's dining experience by endowing food with mobility and conversational capabilities. To demonstrate its interactive capabilities, we have selected four scenarios: Intelligent Dining Guidance, Dietary Behavior Intervention, Food Information Query, and Social Companionship (Fig 1).

4.1 Intelligent Dining Guidance

Based on the optimal ways to consume ingredients, such as the best food combinations and consumption order, the system automatically serves meals to users.

4.2 Dietary Behavior Intervention

The system monitors users' nutritional intake and eating behaviors, providing healthy interventions as needed. For example, if a user has consumed too much of a certain nutrient, the system will refuse to serve foods high in that nutrient to prevent over consumption.

4.3 Food Information Query

Users can look up detailed information about the cultivation, production methods, and nutritional content of ingredients. This feature not only helps users understand the origins and processes behind their food but also adds educational and enjoyable aspects to dining, enhancing their appreciation for the stories behind the food.

4.4 Social Companionship

The system can act as a virtual companion, accompanying users during meals with physical actions and dialogues. This not only alleviates the loneliness of dining alone but also enhances the pleasure of eating through pleasant interactions.

5 DISCUSSION AND FUTURE WORKS

This study introduces the DishAgent system, demonstrating how large language models and intelligent dining table robots can enhance the user's dining experience. By endowing food with mobility and conversational capabilities, the system showcases its strong interactive abilities across four scenarios: Intelligent Dining Guidance, Dietary Behavior Intervention, Food Information Query, and Social Companionship. This not only improves the healthiness and informativeness of dining but also adds a social dimension, significantly enhancing the overall dining experience for users. However, this study has some limitations, and more user experiments are needed to evaluate whether the system can improve the user's dining experience.

Future work will focus on the following aspects: 1) Further optimizing the system's algorithms and hardware design to improve its stability and user experience; 2) Conducting large-scale user tests to collect feedback and iteratively improve the system to ensure it meets the needs of different users.

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